

10 Markov Chain Monte Carlo Methods

The Monte Carlo method is a computer based exploitation of the Law of Large Numbers to estimate a certain probability or expectation. It works like this: Suppose you wish to estimate EY , and that you have an algorithm that will produce a stream of i.i.d. random variables $Y_1, Y_2, \dots$, each with the same distribution as Y . Then the sample average will, at least for large sample sizes n , be a good estimator of the expectation EY . Often, however, in more complex problems it is not so easy to produce i.i.d. copies of Y , and so the naive approach fails.

Markov chain Monte Carlo is a more sophisticated technique. It works like this: Suppose, once again, that you wish to approximate an expectation EY . Suppose that you have an algorithm that will generate successive states $X_1, X_2, \dots$ of a Markov chain on a state space $\mathcal{X}$ with stationary distribution π , then the sample averages of X_i may be used as estimators of EY .

Note that the X_i 's are not independent, as they come from a Markov chain. So we need a Law of Large Numbers for Markov chains. This follows from the Ergodic theorem that we shall not go into.

Markov chain Monte Carlo is useful because it is often much easier to construct a Markov chain with a specified stationary distribution than it is to work directly with the distribution itself. While this sounds paradoxical, it isn't, as the examples below will show. However, the use of estimators based on Markov chains introduces certain complications that do not arise for sample averages of i.i.d. random variables. In the i.i.d. case, the variance of the estimator is just the product of the sample size and the variance of Y . If, on the other hand, the random variables are functions of the successive states of a Markov chain, then there is no longer any such simple rule governing the variances of the sample averages.

Example: Uniform Distribution on Contingency Tables Here is an example which arises naturally in statistical inference (test of independence of attributes) where it is difficult to simulate from the distribution itself, but not so with a Markov chain that has this stationary distribution. Suppose one has an $n \times m$ contingency table. The index pairs (i, j) will be referred to as cells and the element in the (i, j) th cell is $n_{i,j}$. Define the marginals of the table to be the row and column totals; these are denoted by $r_i = \sum_{j=1}^m n_{i,j}$ and $c_j = \sum_{i=1}^n n_{i,j}$. In tests of independence of attributes, the p -value is the probability of a table as extreme or more than the one actually observed, say in terms of the χ^2 statistic, given a particular set r_i and c_j of row and column totals. If the expected frequency in each cell is not large enough, we cannot use the normal approximation. Hence the χ^2 statistic does not have an approximate χ^2 distribution.

Let us denote by T the set of all possible tables with marginals r_i and c_j , and let π be the uniform distribution on T . We wish to simulate the distribution π , compute the χ^2 from each table, find the location of the observed χ^2 in the histogram and the corresponding right tail probability. Even for tables of moderate size (e.g., $n = m = 10$) it may be quite difficult just to list the tables T with given marginals, much less enumerate them. Hence, for large tables, it is virtually impossible to compute the normalizing constant for the uniform distribution on T . (When the row and column totals are all the same, the allowable tables are known as magic squares. In this case it is possible [but not easy] to give an explicit representation of the cardinality of T).

10.1 Metropolis-Hastings Algorithm

It is often the case in simulation problems that the distribution π one wants to simulate has a normalizing constant that is unknown and prohibitively difficult to calculate. Thus, one can write $\pi(x) = Cw(x)$ where w is computable but C is not. Fortunately, it is often possible to build Markov chains with stationary distribution π without knowing C . If, for example, one can find transition probabilities $p(x, y)$ such that the detailed balance equations

$$w(x)p(x, y) = w(y)p(y, x)$$

hold, then they must also hold (**Exercise 1: Show this**) when w is replaced by π .

There is an easy and flexible way to find solutions to the detailed balance equations due to Metropolis. This is the basis for what is now called the Metropolis-Hastings algorithm, perhaps the most common method of simulation now used. The idea is this: Suppose that $\pi(x) = Cw(x)$ is a probability distribution on a finite set $\mathcal{X}$ such that $\pi(x) > 0$ for every x . Suppose that $\mathcal{X}$ is the vertex set of a connected graph G . Choose d large enough that no vertex is an endpoint of more than d edges. For each pair $x, y \in \mathcal{X}$ of distinct vertices, define

$$\begin{aligned} p(x, y) &= \min\left(\frac{w(y)}{w(x)}, 1\right) / d, & \text{if } x, y \text{ share an edge} \\ p(x, y) &= 0 & \text{if } x, y \text{ do not share an edge} \\ p(x, x) &= 1 - \sum_{y \neq x} p(x, y). \end{aligned} \tag{1}$$

Note that $p(x, x) \geq 0$, because $p(x, y) \leq 1/d$ for each pair x, y of neighboring vertices, and no vertex x has more than d neighbors. Thus, equations (1) defines a system of transition probabilities on $\mathcal{X}$. It is easy to check (**Exercise 2: Do it!**) that the detailed balance equations hold for this set of transition probabilities. Since the graph G is connected, the transition probability matrix is irreducible. By choosing d strictly larger than the degree of some vertex, the

transition probability matrix is aperiodic.

Note: You might wonder why we don't just use the complete graph on the vertex set $\mathcal{X}$. The reason is this: In many problems, the weight function $w(x)$ is computable, but not easily so. In such problems it is often advantageous to restrict possible transitions to those pairs for which the ratio $w(y)/w(x)$ is easy to compute. Also, we may not know the number of states, which is required for d , as in the example of the contingency tables.

Claim The transition matrix defined by (1) has π as stationary distribution.
Proof: The steps are

1. Detailed balance equations hold for p defined by (1) with w . (Shown in Exercise 2)
2. Detailed balance equations hold when w is replaced by π . (Shown in Exercise 1)
3. If detailed balance equations hold for π , then π is the stationary distribution. **Exercise 3: Show this.**

10.1.1 Back to the example:

We will use the Metropolis-Hastings method to find a reversible Markov chain on the set T whose stationary distribution is uniform. Observe that, since the target stationary distribution π is uniform, the ratio $w(y)/w(x) = 1$ for all pairs x, y of allowable tables, and so its computation is not an issue here. The difficulty in this problem is in finding a suitable graph structure (that is, building a set of edges connecting allowable tables) without actually having to list the vertices.

Neighboring Tables: Given a table $x = \{x_{ij}\} \in T$, one may obtain another table $y = \{y_{ij}\}$ as follows: choose two rows I, I' and two columns J, J' and set

$$\begin{aligned} y_{IJ} &= x_{IJ} + 1 \\ y_{IJ'} &= x_{IJ'} - 1 \\ y_{I'J} &= x_{I'J} - 1 \\ y_{I'J'} &= x_{I'J'} + 1 \end{aligned}$$

set $y_{ij} = x_{ij}$ for all other cells. Observe that if $x_{IJ} = 0$ or $x_{I'J} = 0$ then the corresponding cells of y will have negative entries. So $y \notin T$. Otherwise $y \in T$, because y has the same marginal totals as x . If table y may be obtained from table x in the manner just described, then tables x and y will be called neighbors. Note that if this is the case, then x can be obtained from y in the same manner. Write $x \sim y$ to denote that tables x, y are neighbors.

The Transition Probability Kernel: A simple Markov chain on the set T may be run as follows: given that the current state is x , choose a pair (I, I') of rows and a pair (J, J') of columns at random; if the neighboring table y is an element of T (that is, if it has no negative entries) then move from x to y ; otherwise, stay at x . The transition probabilities of this Markov chain are given by (1) with $w(x) = w(y)$ and $d = mn(m-1)(n-1)$. Note that the transition rule does not require an explicit description or enumeration of the set T . If these were available, then the uniform distribution on T would, in effect, also be known, and so the use of MCMC would be unnecessary. Note also that the transition rule is computationally simple: it involves only random selection of four integers, and access and modification of the entries in only four cells of the table.

Claim The transition probability matrix P is aperiodic and irreducible on T , and its unique stationary distribution is the uniform distribution on T .

Proof: The transition probabilities clearly satisfy the detailed balance equations $p(x, y) = p(y, x)$, and so the uniform distribution is stationary. That it is the unique stationary distribution will follow from the aperiodicity and irreducibility of the transition probability matrix P .

To prove that the transition probability matrix is aperiodic it suffices to show that there is a state x such that $p(x, x) > 0$. Suppose that $x \in T$ is a table with at least one vacant cell, that is, a cell (i, j) such that $x_{ij} = 0$. If the random selection of I, I', J, J' results in $I = i$ and $J = j$ then the attempted move to the neighboring table y fails, because $y_{ij} = 1$. Thus, $p(x, x) > 0$. Consequently, to show that P is aperiodic it suffices to show that there is a table $x \in T$ with a vacant cell.

Is there a table $x \in T$ with a vacant cell? To see that there is, start with any table x , and set $I = J = 1$ and $I' = J' = 2$. Apply the rule to obtain a neighboring table y . Then apply the rule again to obtain a table y^* neighboring y . Continue this process indefinitely until a table with a vacant cell is reached: this must occur in finitely many steps, because at each step the $(2,1)$ cell is decreased by one.

Finally, to prove that the transition probability matrix is irreducible, it is enough to show that for any two tables $x, y \in T$ there is a finite chain of neighboring tables starting at x and ending at y . That this is always possible is left as Exercise 4.

Exercise 4 Prove that for any two tables $x, y \in T$ there is a finite chain of neighboring tables starting at x and ending at y . Hint: Use induction on the number of rows.

10.1.2 General Form of Metropolis Hastings

Suppose that $\pi(x) = Cw(x)$ is a probability distribution on a finite set $\mathcal{X}$ such that $\pi(x) > 0$ for every x . Our intent is to draw samples from π . Consider any transition matrix $Q = ((q_{xy}))$ on $\mathcal{X}$. The transition probability of the Markov

chain is

$$\begin{aligned} p(x, y) &= \min \left(\frac{w(y)q_{yx}}{w(x)q_{xy}}, 1 \right) q_{xy} \quad \text{for } y \neq x \\ p(x, x) &= 1 - \sum_{y \neq x} p(x, y). \end{aligned} \tag{2}$$

In the previous section, we took a graph on $\mathcal{X}$ and put a probability $1/d$ on each edge of the graph. This was a special case of the Q matrix, with proper weighting on the diagonals to make the row sums one.

When we have a continuous density π , we can use the same procedure with Q being a transition function on the support of π . For example when the support is $\mathbb{R}^k$, then $Q(x, \cdot)$ may be multivariate normal with mean x and a fixed covariance matrix.

10.2 Gibbs Sampler

The Gibbs sampler is a technique suitable for generating a Markov chain with suitable properties that has its stationary distribution in multi-dimensional space. The most interesting aspect is that to run this Markov Chain, it suffices to generate observations from univariate distributions. Suppose the stationary distribution is $\pi(\theta)$ where $\theta = (\theta_1, \dots, \theta_k) \in \mathbb{R}^k$. In Gibbs sampling you proceed as follows.

- Begin with the initial realization $\theta^{(0)} = (\theta_{10}, \dots, \theta_{k0})$
- Generate θ_{11} from the conditional distribution $\pi_1(\theta_1 \mid \theta_{20}, \dots, \theta_{k0})$
- Generate θ_{21} from the conditional distribution $\pi_2(\theta_2 \mid \theta_{11}, \theta_{30}, \dots, \theta_{k0})$
- Generate θ_{31} from the conditional distribution $\pi_3(\theta_3 \mid \theta_{11}, \theta_{21}, \dots, \theta_{k0})$
- Repeat until Generate θ_{k1} from the conditional distribution $\pi_k(\theta_k \mid \theta_{11}, \dots, \theta_{k-11})$
- This gives the first realization $\theta^{(1)} = (\theta_{11}, \dots, \theta_{k1})$
- Continue to get further realizations $\theta^{(2)}, \theta^{(3)}$ etc.

Example: Gibbs Sampler for 1D Ising Model The joint distribution of the one dimensional Ising model with temperature $t > 0$ is given by

$$\pi(\mathbf{x}) \propto \exp \left(\frac{1}{t} \sum_{i=1}^{d-1} x_i x_{i+1} \right) \quad x_i \in \{+1, -1\}$$

To set up the Gibbs sampling, we need to find the conditional distributions $\pi_i(x) = P(x_i = x \mid \text{all other } x_j\text{'s})$

Exercise Show that $\pi_i(x) \propto \exp(\frac{x}{t}(x_{i-1} + x_{i+1}))$ with $x \in \{+1, -1\}$